Quiz 3 — Sample 1

DATA 412/612

Write your answers in an R script. Load the tidyverse and use `stringr`. This sample has questions only.

A few metacharacters you may need:

- `.` matches any character except a newline.
- `^` matches the start of a string.
- `$` matches the end of a string.
- `*` matches zero or more of the preceding piece; `+` matches one or more; `?` matches zero or one.
- `[]` is a character class. `|` means "or". `()` is a capture group.
- `\` escapes a metacharacter, or starts a special class such as `\d` or `\w`.

1. Use `str_c()` on

```
x <- c("Sam", "John")
y <- c("Sampson", "Johnson")
```

   so the result is the single string `Sam Sampson and John Johnson`.

2. First produce a character string whose value is

```
\.?3^!*7+[
```

   (the leading backslash is one character). Then replace that whole string with `0123456789`. Next, start again from the original string and reverse the order of those ten characters, so the value becomes

```
[+7*!^3?.\
```

3. In a character vector, find any seven- or eight-letter word that ends with `sion`, `tion`, or `tory`, and starts with `p`, `f`, or `d` (either case). Check your pattern on

```
c("Passion", "satisfactory", "factory",
  "satisfaction", "devotion", "pa12tion")
```

4. Find any string that starts with any two letters, then a comma, then at least one space and at most two spaces, and ends with any five digits. Check your pattern on

```
c("MD 21228", "WA, 76731", "VA: 34124", "FL, 21225", "ca, 45691")
```

5. Find any string in which at least three consonants are in a row. Treat this as case-insensitive. Check your pattern on

```
c("HHHonorsion", "DBBELIcious", "Prio5ity", "HIROSHIMA",
  "Uinicasion", "Cat", "Catsion", "Dog", "FininGdg",
  "Extension", "SsSktinger")
```

6. Find any string in which at least three consonants are in a row, where the first and third of those consonants are uppercase and the second is lowercase. If the run is longer than three consonants, do not restrict the case of the later ones. Check your pattern on

```
c("H2norsion", "DELIcious", "Priority", "HIROSHIMA",
  "Uinicasion", "CATti", "CASIONnNbd", "Dog", "Fishing",
  "Extension", "Singer")
```

7. Suppose a word starts with a consonant. Switch that first letter with the first vowel in the word. Apply this to

```
c("DELIcious", "Priority", "HIROSHIMA", "Dog",
  "Fishing", "Extension", "Singer")
```

The next three items are harder pattern problems from the original sample. The real quiz will not require lookahead or lookbehind.

8. Find any word that starts with a capital letter, does not end with `ing`, `ity`, or `sion`, and has at most nine letters. Check your pattern on

```
c("H2norsion", "DElicious", "Prio5ity", "HIROSHIMA",
  "Uinicasion", "Cat", "Catsion", "Dog", "Fishing",
  "Extension", "Singer")
```

9. Give a regular expression that validates an email with these rules: at least one uppercase letter, one lowercase letter, and one digit; the local part uses only `[a-zA-Z0-9_]`; the domain is `gmail.com` or `yahoo.com`. There is no limit on the length of the local part. Check it on

```
c("sam34A23@gmail.com", "Su1ut49@yahoo.com",
  "Su1@ut49@yahoo.com", "Su1-ut49@yahoo.com",
  "jer71d@american.edu")
```

10. Give a regular expression that validates a password-style id with these rules: at least two uppercase letters, exactly one lowercase letter, at least three digits, and at most 10 characters in the id. Check it on

```
c("sA34A2376@gmail.com", "SuBB149@yahoo.com",
  "Su1@ut49@yahoo.com", "Su1-ut49@yahoo.com",
  "JEr78_1U@yahoo.com")
```